THE CLOSURE OF AN ASSOCIATED POINT THAT MEETS THE CLOSED FIBRE MEETS AN ASSOCIATED POINT OF THE FIBRE

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file spaces-flat.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: a lift of the specialisation along an etale chart of a decent space, an identification of the relative assassin through fibre associated points, and an application of the schemes case

Editorial note. The label lemma-associated-point-specializes names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Allowed standard prerequisites (editorial).

- The relative assassin of a finite type module over the base, and its description as the set of points whose fibre point is associated to the fibre of the module
- Etale charts of a decent space and the lifting of specialisations from the space to the chart
- The schemes version of the statement, identifying the associated points of the special fibre of the pullback with the fibre of the relative assassin

Source reference. Original lemma 03IL: sources/stacks-chow/tags.txt, source lines 4607–4607. Prerequisite pointer invoked at proof line(s) 1015; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 0AC8: sources/stacks-chow/tags.txt, source lines 13046–13046. Prerequisite pointer invoked at proof line(s) 1011; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 05L3: sources/stacks-chow/tags.txt, source lines 7222–7222. Prerequisite pointer invoked at proof line(s) 1022; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Lemma 0.1. *Let S be a scheme. Let X be a decent algebraic space locally of finite type over S. Let $\mathcal{F}$ be a finite type, quasi-coherent $\mathcal{O}_X$-module. Let $s \in S$ such that $\mathcal{F}$ is flat over S at all points of X_s. Let $x' \in Ass_{X/S}(\mathcal{F})$. If the closure of $\{x'\}$ in $|X|$ meets $|X_s|$, then the closure meets $Ass_{X/S}(\mathcal{F}) \cap |X_s|$.*

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Proof. Observe that $|X_s| \subset |X|$ is the set of points of $|X|$ lying over $s \in S$, see Decent Spaces, Lemma 0AC8. Let $t \in |X_s|$ be a specialization of x' in $|X|$. Choose an affine scheme U and a point $u \in U$ and an étale morphism $\varphi : U \to X$ mapping u to t. By Decent Spaces, Lemma 03IL we can choose a specialization $u' \rightsquigarrow u$ with u' mapping to x'. Set $g = f \circ \varphi$. Observe that $s' = g(u') = f(x')$ specializes to s. By our definition of $\mathrm{Ass}_{X/S}(\mathcal{F})$ we have $u' \in \mathrm{Ass}_{U/S}(\varphi^*\mathcal{F})$. By the schemes version of this lemma (More on Flatness, Lemma 05L3) we see that there is a specialization $u' \rightsquigarrow u$ with $u \in \mathrm{Ass}_{U_s}(\varphi^*\mathcal{F}_s) = \mathrm{Ass}_{U/S}(\varphi^*\mathcal{F}) \cap U_s$. Hence $x = \varphi(u) \in \mathrm{Ass}_{X/S}(\mathcal{F})$ lies over s and the lemma is proved. $\square$